

EDUCATION

Carnegie Mellon University
Bachelor of Science in Computer Science, Minor in Machine Learning

Pittsburgh, PA
Dec 2026

Coursework: ML (PhD), RL (PhD), Deep Learning Systems (MS), ML with Large Datasets (MS), Distributed Systems

Honors/Activities: SCS Dean’s List, CMU AI Safety Initiative, Scotty Labs

WORK EXPERIENCE

Notion | Software Engineer InternMay 2026 – Aug 2026

- Built hardened sandbox infrastructure for AI agents handling 3k+ sessions/month, with dedicated K8s clusters, interactive and daemon runtimes, per-box egress allowlists, and a MITM proxy for credential injection and I/O sanitization
- Developed a multi-agent system to autonomously scan and patch Notion infrastructure, fixing 57 vulnerabilities to date

Tesla (Autopilot) | Software Engineer InternJan 2026 – May 2026

- Built isolated ML infrastructure (dedicated compute, scheduler, storage, performance tuning) to compile release-bound Autopilot/Optimus neural networks into AI3/4 inference artifacts; serves 8M+ vehicles globally since 2026.8.3 
- Accelerated ML compilation and testing pipeline with weights deduplication (-25% build volume), SDK caching (-20% time)
- Improved model encryption memory footprint by 98%, and reduced throttling during Autopilot simulation runs by 47%
- Migrated 34 services such as orchestrators for inference clusters, simulation, replay across K8s, VM, bare metal to new DCs

Amazon Web Services | Software Engineer Intern, AI/MLMay 2025 – August 2025

- Built service within AWS Agentic AI to translate Bedrock Agents into equivalent agents in other frameworks (eg. Langchain)
- Recreated 20+ Bedrock primitives including guardrails, Lambda tools, memory, agent collaboration, and code sandboxing
- Achieved 95% agent similarity with continuous eval pipeline on feature-conditional prompt suite and 14k randomized agents
- Featured on AWS Console , launched with SDK integration , and validated with clients including Autodesk and Netflix

Hadrian | Software Engineer Intern, AI/MLMay 2024 – Aug 2024

- Developed Python/Go backend and Kafka pipeline (<50ms latency) to enable robotic tool-loading and sync with factory state
- Trained and deployed on-device ML model achieving 98.5% accuracy in fully automated validation of tool position/type
- Cut tool-loading times by 75%, saving 11k technician-hours yearly in building parts for clients such as Anduril and Raytheon

Compos Mentis | Founding EngineerAug 2023 – May 2024

- Developed platform to accelerate case qualification and legal review using LLMs, generating \$90k revenue from 23 law firms
- Fine-tuned LLMs via LoRA and built extraction and RAG pipeline to surface case insights, speeding up analysis by 8x

Machina Labs | Software Development InternJul 2022 – Sep 2022

- Created ML pipeline to improve part alignment using laser scans and Bayesian optimization; deployed with Azure Kubernetes
- Achieved 40% faster iteration and 25% reduced margin of error on parts for clients such as NASA, SpaceX, and the USAF

PROJECTS

RLVR-Predictor Predicts post-RLVR lift before training using a LightGBM model trained on GRPO rollout features; achieved 78% batch-ranking accuracy, 34% faster NLL convergence in fine-tuning Qwen 3 on H100s

Quantum NNs CUDA-accelerated QNN modules, ops with full autograd support for DL framework , outperformed classical NNs by 21-42% in 3 entanglement detection tasks by exploiting phase info

Train of Thought Multi-agent LLM system capable of chain-of-thought reasoning, backtracking with Google Gemini

File-Caching Proxy Low-latency distributed system for remote file operations via a server and LRU chunk-caching proxy

ACHIEVEMENTS

VEX Robotics World Championship (2x) | Achieved top ~30 out of 11k teams as Team Captain (2019-23)

AWS Certified Solutions Architect | **Lockheed Martin Scholar (\$40k)** | **Edison International STEM Scholar (\$50k)**

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Java, Go, SQL, Bash

Infrastructure: AWS, GCP, Azure, Kubernetes, Docker, Kafka, Spark, Airflow, gRPC, Terraform, Redis, PostgreSQL

ML Systems: PyTorch, TensorFlow, JAX, CUDA, Slurm, Ray, vLLM, NumPy, OpenCV, Polars, Langchain, W&B